EWB COURSES

AI-POWERED CYBERSECURITY

Assessment 01

Security Foundations

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Batch / Group	1
Assessment	Individual work | 100 marks
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Report note: This report is based on the fictional message and sample sign-in records supplied in the assessment. No real account, website or person was investigated.

Tasks covered

- Task A — Secure a student account
- Task B — Analyse a suspicious message
- Task C — Triage a sign-in alert
- Task D — Review an AI security answer

Task A — Secure a student account (25 marks)

A1–A2. Main risks, possible impact and suitable controls

I reviewed the situation and listed the main risks to the student's account, laptop and coursework. The controls below can reduce the risks, but no single control can prevent every type of attack.

Risk	What could be affected	Possible impact	Control	Limit
Using the same password for email and the learning portal	Email, portal, assignments and account details	If one password is exposed, someone may get into both accounts and view or change information.	Use a different, long password for each account. A trusted password manager can help store them.	A password manager account can also be targeted. Unique passwords do not stop phishing or a breach at the service itself.
Approving sign-in prompts that were not expected	Account access, messages and personal information	An unknown person may get access and read information or change account settings.	Deny unexpected prompts. If they continue, report them to official IT support. Never share an OTP.	MFA can still be affected by prompt-bombing, phishing, stolen sessions or weak recovery methods.
Delaying device updates	Laptop, saved files, browser sessions and credentials	Attackers may use known software weaknesses to access or damage files.	Turn on automatic security updates and restart when needed after saving work.	Updates do not fix every weakness and can sometimes cause compatibility issues. Backups are still needed.
Keeping the only project copy on one laptop	Project files and coursework	A damaged, lost or infected laptop could make the work unavailable.	Keep a separate backup, such as approved cloud storage or an external drive. Check that files can be restored.	A synced copy may also copy accidental deletion or ransomware changes. Keep version history or a separate backup.
Letting a friend use the laptop while accounts are signed in	Open accounts, private files and coursework	The friend could accidentally or intentionally view, change or delete information.	Lock the screen when away. Sign out and create a separate standard user account for the friend.	A separate account reduces casual access but does not protect against every device-level threat.

Priority order

1. Secure the accounts first. Deny unexpected MFA prompts, contact official IT support if they continue, and change any reused password from a trusted device. This is the first priority because an unauthorised sign-in could expose more than one account.

2. Back up the project. Make a separate copy and check that at least one file can be restored. This helps if the laptop is lost, damaged or affected by ransomware.

3. Update and lock the laptop. Install supported updates, turn on screen locking and use separate user accounts. These steps lower the risk, but they cannot remove every threat.

A3. Authentication and authorisation

Authentication checks who is signing in. For example, Mira enters her account details and completes a second step to sign in to the college portal. Authorisation is about what she can do after signing in. She may be allowed to view and submit her own assignments, but she should not be able to change another student's marks or open staff-only records. In simple words, authentication asks “Who are you?” and authorisation asks “What are you allowed to do?”.

Task B — Analyse a suspicious message (25 marks)

B1. Warning signs in the supplied message

I would treat this message as suspicious because it asks the student to act quickly and share information that should remain private.

What the message says	Why it is a warning sign
Sender is awards@campus-awards.example but Reply-To is help@personal-awards.example	The sender and reply addresses use different domains. This needs to be checked using contact details found separately from the message. The mismatch alone is not final proof.
“FINAL WARNING” and “expires in 20 minutes”	The short deadline creates pressure and may stop the student from checking whether the message is real.
It asks for the college password and OTP	A password and one-time code should not be shared through a message or an unknown link.
“Do not contact the college office”	This discourages the student from checking the request with the official college team. That is suspicious.
It asks for a photo of the identity card	An identity card contains personal information. It should not be sent unless the request and secure process are verified.

B2. How I would verify the message safely

1. I would not click the link, reply to the message, enter a password or OTP, or send an identity-card photo.

2. I would open the college portal using an address I already know, or use contact details from official college material. I would not use the link or phone number in the suspicious message.

3. I would contact the college or scholarship office and ask whether the award and verification request are genuine. I would also report the message through the official channel.

4. I would remember that HTTPS only protects the connection between the browser and website. It does not prove that the website belongs to the college. A familiar sender name can also be misleading, so the full domain and request need to be checked.

B3. If the student has already entered a password and OTP

1. From a trusted device, change the password on the real college account straight away. Change it anywhere else it was reused. Do not share the new password or any OTP.

2. Contact the college IT or security team using independently verified contact details. Explain what happened and ask them to review the account and end active sessions.

3. Use the account's security settings to sign out other sessions, check MFA methods and recovery details, and remove unknown devices or app access. Follow the IT team's advice.

4. Report the suspicious message and keep the original message and its details for review. Do not send identity documents around unnecessarily.

5. Check recent sign-ins, forwarding rules, recovery settings and portal activity. Report anything unfamiliar.

B4. What cannot be confirmed from the message alone?

The message by itself cannot tell us who really sent it, who controls the link, or whether an account has already been accessed. To investigate further, the college could check the full email headers, domain information, official scholarship records and portal audit logs. The office should also confirm the request through a separate, trusted contact method. The .example addresses in this task are fictional training details, so they do not prove that a real sender or website is malicious.

Task C — Triage a sign-in alert (25 marks)

C1. Timeline: what the logs show and what it may mean

Time (IST)	What is shown in the sample log	My interpretation
09:00–09:01	Password sign-in succeeded from 192.0.2.10 and MFA was approved. The lab notes call this the usual classroom network address.	This looks consistent with the student's normal sign-in. Still, a log entry alone cannot prove who was using the account.
10:14–10:16	Two password failures came from 198.51.100.23, followed by a successful password sign-in from the same address.	This could be someone guessing the password, or the student entering it incorrectly before succeeding. It needs checking. The IP address does not identify a person.
10:17–10:18	MFA was denied and then approved. The student says they approved a prompt because they thought it would stop repeated requests.	The approval may have allowed an unwanted sign-in. The student's explanation is important, but the log cannot show who held the device or why the prompt appeared.
10:20	A session was created after the password success and MFA approval.	The sign-in sequence appears to have created a session. The IT team should check whether it is still active and what happened after it was created.

C2. Possible explanation, alternative and checks

Possible explanation: Someone else may have known or guessed the student's password. After the student approved an unexpected MFA prompt, that person may have been able to create a session. The password failures, later success, MFA denial and approval, and session creation make this a possible account compromise that should be investigated.

Another possibility: The student may have been using a legitimate service, VPN or network that showed a different network address. They may also have denied one prompt by mistake and approved the next one. The student says they did not sign in after 09:05, so this explanation needs to be checked rather than assumed.

Two useful checks: First, review the identity-provider audit records for device and browser details, session ID, MFA method and any activity after 10:20. Second, ask the student and authorised IT staff whether a VPN, known device, password reset or legitimate service could explain the event. The IP address should only be treated as a clue, not proof of a person's identity.

C3. Three actions for the authorised IT team

1. Secure the account. Revoke active sessions and refresh tokens, reset the password through the official process, review MFA and recovery settings, and block suspicious access if policy allows. Coordinate with the student so the account is secured without causing an unnecessary lockout.

2. Preserve the evidence. Save the relevant password, MFA and session logs with timestamps and time zone. Keep related alerts and endpoint records, record who collected them, and limit access. Do not delete the suspicious logs.

3. Check how far the activity went. Review actions after the session was created, email forwarding rules, recovery changes, connected apps and any other affected accounts. Remove unauthorised changes, confirm the account is safe and record the follow-up.

C4. Incident summary

On 18 October 2026, student07 signed in successfully and approved MFA from the usual classroom network at about 09:00. Later, two password attempts failed from 198.51.100.23, followed by a successful sign-in. An MFA request was denied and then approved, and a session was created at 10:20. The student says they did not sign in after 09:05 and approved a prompt to stop repeated requests. A possible account compromise needs urgent checking, but the IP address does not identify a person.

Task D — Review an AI security answer (25 marks)

D1. Five unsafe or unsupported claims and my corrections

AI answer or action	My correction
“The email is safe because it uses HTTPS.”	This is not a safe conclusion. HTTPS encrypts the connection, but it does not prove that the website is genuine. The urgent deadline, different reply address and requests for a password, OTP and identity-card photo still need to be treated as warning signs.
“The logs prove that the person using 198.51.100.23 stole the password.”	The logs show activity linked to that sample IP address. They do not identify a person or explain how the password was obtained. A VPN, network change or user error should also be checked.
“MFA prevents every account compromise.”	MFA lowers risk, but it cannot prevent every attack. A person may approve a prompt by mistake, or an attacker may use phishing, a stolen session or a weak account recovery process. Unexpected prompts should be denied and reported.
“Delete the suspicious logs.”	This is unsafe because logs may be needed to understand what happened. The authorised team should preserve the records with their timestamps and control who can access them.
“Send the student's password to support so they can test it.”	Support staff should never need the student's password or OTP. The account should be secured through an official password reset and session-revocation process. Passwords and codes must remain private.

D2. A safer prompt for an AI tool

> “Review only the fictional message and sample sign-in logs provided in this assessment. Separate the facts shown in the evidence from possible explanations and things that are still unknown. Point out the security risks and suggest safe actions. Do not say that an IP address identifies a person or that HTTPS proves a website is trustworthy. Do not ask for passwords, OTPs, identity documents or real personal information. Explain uncertainty and mention what further authorised evidence would help. Do not open the fictional links or test any live system.”

D3. Corrected security advice (100–150 words)

I would not trust this scholarship message just because it uses HTTPS or shows a familiar college name. It creates urgency, uses a different reply address, asks for a college password and OTP, tells the student not to contact the office, and requests an identity-card photo. The student should not click the link or reply. They should check the scholarship through the official college portal or contact the office using details found separately. The sign-in log shows a password success, an MFA denial, an MFA approval and a new session, but it does not show who was behind the IP address or how the password was obtained. The student should report the message, and the IT team should secure the account and preserve the logs.

D4. AI-use declaration

ChatGPT was used to help organise the report, explain some security concepts and review the fictional examples in the assessment. I reviewed the content and am responsible for checking that the final report is accurate and that I understand the answers. The assessment examples are fictional. No live systems were tested, no links from the fictional message were opened, and no real passwords or OTPs were used.

Evidence notes

CISA — Secure Our World: I used this guidance for general advice about unique passwords, multifactor authentication, software updates and recognising or reporting phishing.

NIST — AI 600-1: This guidance explains risks in generative AI, including incorrect or made-up information. It supports checking AI answers against the evidence instead of accepting every statement as correct.

References

1. Cybersecurity and Infrastructure Security Agency (CISA). *Secure Our World*. https://www.cisa.gov/secure-our-world

2. National Institute of Standards and Technology (NIST). *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*, NIST AI 600-1. https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

3. EWB Courses. *AI-Powered Cybersecurity — Assessment 01: Security Foundations*. Assessment brief issued 18 October 2026.

Final review before uploading

- Check that the name and Batch 1 details are correct.
- Read the answers and make sure you can explain the points in your own words.
- Check that the sign-in log section separates what was observed from what is only a possible explanation.
- Keep the AI-use declaration accurate and disclose any additional tools used.
- Follow the Google Form instructions shared by EWB and use the deadline announced by the team.

This report discusses only the fictional evidence supplied for the assessment. It does not claim that a real email, account, IP address or website has been investigated.